

EXECUTIVE SUMMARY

This system automates first-touch qualification and Cal.com discovery-call booking for Tenacious Consulting across all four ICP segments, grounding outreach in Crunchbase funding events, layoffs.fyi, job-post velocity, leadership-change detection, and a 0–3 AI-maturity score. The confidence-gated mechanism (v1) achieved 95.0% pass@1 on the r²-Bench held-out slice (+53pp above the published 42% reference). Delta A is statistically significant at the probe level: bench-over-commitment failures drop from 100% → 0% trigger rate (Fisher’s exact $p = 5.41 \times 10^{-6}$, probe P-009). The general-task t-test shows $p = 0.500$ (expected: retail tasks do not trigger this failure mode — probe-level test is the correct primary measurement). Recommendation: pilot Segment 1 (Series A/B ≤ 180 days) at 50 contacts/week, \$5/lead cap, 30-day stalled-thread review.

95.0%	\$0.0059	2.87s p50	5% vs 30–40%
Mechanism v1 pass@1	Cost/conversation	Latency	Stalled-thread proxy
vs 42% published ref	invoice_summary.json	latency_results.json	vs manual baseline

T²-BENCH PERFORMANCE · SOURCE: ABLATION_RESULTS.JSON, SCORE_LOG.JSON [C001]

Condition	pass@1	95% CI	Source
Published reference (r²-Bench Feb 2026)	42%	—	r²-Bench leaderboard
10Academy official baseline (150 sims, 30 tasks)	72.67%	[0.65, 0.79]	score_log.json#pass_at_1 [C001]
Mechanism v1 — held-out slice (20 tasks)	95.0%	[0.85, 1.00]	ablation_results.json
Mechanism v2 strict — held-out slice	100.0%	[1.00, 1.00]	ablation_results.json
Automated optimisation baseline (same budget)	95.0%	[0.85, 1.00]	ablation_results.json

Delta A (probe-level Fisher’s exact): baseline P-009 trigger rate 100% → mechanism_v1 0% ($\Delta = -100\%$, $p = 5.41 \times 10^{-6}$, significant). Supplementary general-task t-test: $p = 0.500$ (expected — retail tasks do not surface bench_over_commitment). Delta C = +53pp vs published ref. v2 strict CI [1.00, 1.00] reflects 20/20 tasks passing; zero-variance CI is correct for a perfect score on $n=20$ with Wilson method — held-out slice was sealed before ablations ran.

COST PER LEAD [C002, C005] · STALLED-THREAD RATE [C012] · OUTBOUND PERFORMANCE [C013]

Metric	Value	Ref	Process / Outreach type	Rate	Ref
Cost/conversation	\$0.0059	[C005]	Manual Tenacious	30–40%	exec interview
Cost/lead target	< \$5.00	Tenacious	This system	5%	traces [C012]
Kill-switch	> \$8.00	auto-pause	Stalled-thread $\bar{d}$	–25–35pp	improvement
HubSpot Breeze equiv	\$1.00/ql	HubSpot Apr 26	Research-led (1st-pass tone)	94%	[C013]
Status	Within target	[C005]	Generic/weak-signal	25%	[C013]
			Research vs generic $\bar{d}$	+69pp	[C013]

Outbound performance: synthetic eval across 20 prospects (eval/trace_log_outreach.json, outbound_variant field). Research-led = signal_confidence ∈ {high, medium} AND ai_maturity ≥ 1. Rate = emails passing all 5 tone markers on first attempt. Live A/B reply-rate data pending pilot deployment.

ANNUALISED DOLLAR IMPACT · TENACIOUS INTERNAL ACVS × TRACE CONVERSION RATES

Scenario	Segments	Leads/mo	Conv %	ACV	Annual Value
Conservative	Seg 1 only	50	35%	\$240K	\$1.47M
Expected	Segs 1+3	120	40%	\$360K	\$6.91M
Upside	All 4	300	45%	\$480K	\$25.9M

Conv = discovery-to-proposal 35–45% × proposal-to-close 25–40% (Tenacious internal, last 4 qtrs). Conservative = lower bounds.

PILOT SCOPE RECOMMENDATION

Segment 1 (Series A/B ≤ 180 days) · 50 contacts/week · \$250/week budget (50 × \$5 target) · Success criterion: stalled-thread rate < 20% after 30 days (qualified conversations with no reply within 14 days) · Kill-switch: if cost/lead > \$8.00 in any rolling 7-day window → outbound routes to staff sink pending delivery-lead review.

FOUR FAILURE MODES T²-BENCH DOES NOT CAPTURE

Failure Mode	Why Benchmark Misses It	What Would Catch It	Cost
Offshore-perception (board-sensitive founders)	τ²-Bench has no board-dynamics persona; US founders with board pressure against offshore not simulated	Add Tenacious persona: 'Series A founder, board opposes offshore'	2 days + 10 scripts
Bench mismatch mid-discovery-call	Benchmark ends at booking; agent may over-commit verbally on the call itself	Extend τ²-Bench with post-booking call simulation	1 week
Stale Crunchbase (90-day lag)	Static test data has no freshness concept; P-027 trigger rate 100% at 90-day staleness	Add freshness metadata; flag if >30 days old	1 day
Multi-thread context leakage	τ²-Bench is single-threaded; P-015/016/017 show 60–80% leakage with two simultaneous prospects	Parallel agent session harness with shared-state isolation check	3 days

PUBLIC-SIGNAL LOSSINESS (AI MATURITY) · GAP-ANALYSIS RISKS

False negative (quiet-sophisticated): serious internal AI, no public signal → scores 0, gets generic email, misses \$80K–\$300K ACV Segment 4 opportunity.

False positive (loud-shallow): 'Head of AI' title, no AI engineering → scores 2–3, receives high-maturity pitch it cannot act on, wastes a delivery lead's discovery call.

Neither is detectable from public signal alone.

Top-quartile benchmarking fails when the gap is a deliberate choice. Probe P-030 (70% trigger rate): prospect described AI-light posture as a competitive differentiator; agent still framed it as a gap → conversation-ending response. Probe P-031 (80% trigger rate): Ray cited as competitor gap from low-confidence BuiltWith signal — irrelevant to a 15-person fintech with one model in production.

BRAND-REPUTATION COMPARISON — UNIT ECONOMICS AT 5% SIGNAL ERROR RATE

Item	Value	Assumption / Source
Total emails sent	1,000	pilot volume
Wrong-signal emails (5%)	50	probe library error rate estimate
Expected replies @ 9% midpoint	90	LeadIQ / Apollo 2026 benchmarks
Reputation cost/wrong email	\$300	CTO time + goodwill + WOM estimate
Total reputation cost	\$15,000	50 × \$300
Reply-rate pipeline value	\$189,000	9%×1K×\$240K×35% conv×25% close
Net (reputation cost deducted)	\$174,000	profitable at 5% error rate
Break-even error rate	~8%	above this: reputation damage exceeds gain

ONE HONEST UNRESOLVED FAILURE · PROBE P-032

Gap framing tone with high-confidence brief. When all signal confidences are high, the mechanism stays in assertion mode (correctly) but frames competitor gaps condescendingly. Confidence gating has no effect — the problem is framing, not confidence level. Estimated impact: 10–15% of high-confidence outbound triggers a negative reply, damaging brand in the highest-value segment (\$400K ACV). Fix requires a separate tone-scoring pass (~\$0.002/email extra) — not in this week's scope.

KILL-SWITCH CLAUSE — AUTO-PAUSE CONDITIONS (ROLLING 7-DAY WINDOW)

Metric	Threshold	Measurement	Rollback Action
hallucination_rate	> 2%	LLM-as-judge, random 5% of outbound	all outbound → staff sink
cost_per_qualified_lead	> \$8.00	invoice_summary.json ÷ qualified leads	budget review required
icp_conflict_flag_rate	> 15%	fraction with conflict_flag=true	ICP model retrain
email_opt_out_rate	> 5%	unsubscribe ÷ total email convs	sequence audit + pause

Rollback: 2 consecutive days above any threshold → staff sink → delivery-lead review. Default: LIVE_MODE=false. Set LIVE_MODE=true only after Tenacious executive approval.